# Sonia Irakoze

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## EDUCATION

**University of Rochester, Hajim School of Engineering & Applied Sciences** — **Rochester, NY**
*Bachelor of Science in Mechanical Engineering* — *Expected May 2028*

- Alan & Jane Handler Endowed Scholarship (top 1% merit award) • Kearns Scholar • Dean's List (Fall 2024)

## EXPERIENCE

**Colorado Department of Transportation (CDOT), Staff Bridge Branch** — **Denver, CO**
*Engineering Intern, Bridge Asset Management* — *Summer 2026*

- Drafted and revised bridge rehabilitation linework and plan-view sketches in OpenRoads Designer; managed files in ProjectWise
- Wrote SQL queries (SSMS) on the SIMSA bridge database; reconciled records against OnBase, OTIS, and project files
- Reviewed plans and pay items for 16 Region II projects, logging affected structures and proposed treatments in a tracking dataset
- Built **CDOT Compass**, a React/TypeScript web tool mapping CDOT for new staff; named Headquarters Division Winner for Outstanding AI Improvement in the 2026 CDOT Innovations Challenge

**AGR Sensors** — **Rochester, NY (Remote)**
*Data Research Intern, CropSight AI Pipeline* — *January 2026 – Present*

- Audited 250+ source chunks from 22 agricultural reference books for CropSight, an AI crop-disease diagnosis tool
- Built Python validation checks for 212 disease and pest entries catalogued as ML-ready JSON, reaching zero schema errors

**Ugandan Water Project** — **Lima, NY**
*Development Intern, Schwartz Community-Based Internship Scholar* — *June 2025 – August 2025*

- Contacted 100+ donors and corporate sponsors in targeted outreach, helping raise $5K+ for clean water projects in Uganda
- Cleaned and organized CRM donor records for engagement tracking and ran communications for the annual golf tournament

**University of Rochester IT Center** — **Rochester, NY**
*Level 2 Training Coordinator (Student Lead); promoted from Help Desk Technician* — *January 2025 – Present*

- Resolve escalated hardware, network, NetID/Duo login, and malware cases in Jira for students, faculty, and staff
- Train and mentor new Level 1 student technicians on troubleshooting procedures, ticket escalation, and service standards

## PROJECTS

**SAquaSolve Rwanda** — **Gashora, Rwanda**
*Founder & Project Lead* — *August 2023 – Present*

- Founded and lead SAquaSolve Rwanda, growing a grant-funded student initiative into an emerging Rwandan nonprofit; direct strategy, governance, budgeting, operations, and NGO registration
- Directed a 32-page pre-feasibility study reviewed by practicing water engineers, advancing a single-kiosk concept into a solar-powered intake, treatment, and distribution system within a $145,000 implementation plan
- Secured a competitive $10,000 Davis Projects for Peace grant; lead fundraising strategy, proposals, and budgets while building relationships with Rotary clubs, government representatives, and water-sector organizations
- Spearheaded a youth-led WASH initiative, mobilizing 53 participants and establishing a Youth Ambassador program to drive community engagement, education, and local leadership

**Engineering Design Projects, University of Rochester** — **Rochester, NY**
*Design Lead, Aero Design; Structures Team Member, CubeSat; Bridge Design Coursework* — *2024 – 2026*

- *Aero Design:* Compared wing and tail designs with FEA and built laser-cut prototypes over 15+ iterations; team cut weight ~10%
- *CubeSat:* Modeled satellite frame parts in Siemens NX and simulated launch vibration and thermal loads in MATLAB
- *Bridge Design:* Reduced modeled weight of a 52 m timber truss 34% from my first design within buckling and deflection limits

## LEADERSHIP

Steering Committee Member, University of Rochester For Ever Better Campaign (one of two students selected) • NSBE Upstate Zone Publications Chair (2026–27) and Chapter Senator (2025–26) • UniBooth Co-founder (won iZone Make It Happen Prize)

## AWARDS

U.S. State Dept. TechGirls (one of 3 from Rwanda, 2023) • 3rd Prize, East African Community Essay Competition (2022) • Silver Award, Queen's Commonwealth Essay Competition (2023) • Best Entrepreneurial Vision, Heist Hackathon (MarketBite, 2026)

## SKILLS

**Engineering:** Siemens NX, OpenRoads Designer, ProjectWise, Fusion 360, FEA, GD&T; lathe, mill, 3D printing, laser cutting
**Software & Design:** SQL, Python, MATLAB, Git, HTML/CSS, React/TypeScript; Adobe InDesign, Illustrator, Figma, Canva
**Languages:** Kinyarwanda (native), English (fluent), French (intermediate)